PRE-ALGEBRA (8TH) · UNIT 1: REAL NUMBERS AND EXPONENTS

Space Mission Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan a crewed mission to Mars. They compare enormous distances in scientific notation, compute travel times, radio delays and food needs, and size habitat modules with roots and exponent rules.

The project ties scientific notation, integer exponents and irrational numbers to one real decision: can the mission be done?

Standards Addressed

- 8.NS.A.1 Rational and irrational numbers
- 8.NS.A.2 Approximate irrational numbers
- 8.EE.A.1 Properties of integer exponents
- 8.EE.A.2 Square and cube roots
- 8.EE.A.3–4 Scientific notation and operations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the distance cards. Part 1: writing and comparing numbers in scientific notation.
Day 2	Part 2: travel time, radio delay, and supplies.
Day 3	Part 3: habitat design with roots and exponents; mission poster or briefing.

Materials

- Student handout with the Mission Data Cards
- Calculators with scientific notation
- Poster paper
- Optional: NASA Mars mission video clip

How the Three Versions Differ

Level	What changes
CP2	Guided scientific notation with starters, single-step division for time, perfect squares and cubes.
CP1	Multi-step comparisons, unit conversions, negative and zero exponents, and estimating irrational roots.

Level	What changes
Honors	Interstellar distances, required-speed problems, exponent proofs, and an open food-budget design.

Watch For (Common Misconceptions)

- Adding exponents when dividing powers of 10 instead of subtracting.
- Leaving the first factor outside 1–10 (for example 0.96×10^{1}).
- Thinking $\sqrt{50}$ is rational because 50 is a whole number.

Extension Ideas

- Research a real Mars mission timeline and compare it with your travel time.
- Find the radio delay to Jupiter and Saturn.
- Model the habitat with cubes and compute surface area for insulation.

Answer Key – CP2

Part 1: Big Numbers

#	Task	Answer
1	Write the distance from Earth to the Moon, 384,000 km, in scientific notation.	3.84×10^5 km
2	Write 5.46×10^7 km in standard form.	54,600,000 km
3	About how many times as far is Mars (closest) as the Moon? Divide and round to the nearest whole number.	$(5.46 \div 3.84) \times 10^{7-5} \approx 1.42 \times 10^2 \approx 142$ times as far.

Part 2: Travel Time

#	Task	Answer
4	The rocket travels 4×10^4 km per hour. How many hours does it take to reach the Moon?	$(3.84 \times 10^5) \div (4 \times 10^4) = 0.96 \times 10^1 = 9.6$ hours.
5	How many hours does the rocket take to reach Mars at its closest? About how many days is that?	$(5.46 \times 10^7) \div (4 \times 10^4) = 1.365 \times 10^3 = 1{,}365$ hours ≈ 56.9 days.
6	A radio signal travels 3×10^5 km per second. How many seconds does a message take to reach Mars at its closest?	$(5.46 \times 10^7) \div (3 \times 10^5) = 1.82 \times 10^2 = 182$ seconds (about 3 minutes).

Part 3: Habitat Design

#	Task	Answer
7	A cube-shaped sleeping module has a volume of 512 cubic meters. How long is each edge?	$\sqrt[3]{512} = 8$ meters.
8	A square solar panel has an area of 49 square meters. How long is each side?	$\sqrt{49} = 7$ meters.
9	Each supply crate holds 2^3 boxes, and each box holds 2^4 meal packs. Write the number of meal packs in one crate as a power of 2 and as a number.	$2^3 \cdot 2^4 = 2^7 = 128$ meal packs.
10	Is $\sqrt{50}$ rational or irrational? Between which two whole numbers is it?	Irrational (50 is not a perfect square). $7^2 = 49$ and $8^2 = 64$, so $\sqrt{50}$ is between 7 and 8 (about 7.07).

Reflection

#	Prompt	Look for
R1	Why is scientific notation useful for space distances?	It writes very large numbers in a short form and makes them easy to compare and divide using exponents.

#	Prompt	Look for
R2	How do you divide numbers in scientific notation?	Divide the decimal parts and subtract the exponents of 10, then adjust so the first part is between 1 and 10.

Answer Key – CP1

Part 1: Distances & Comparisons

#	Task	Answer
1	Find the difference between the Mars (closest) distance and the Moon distance. Write the answer in scientific notation.	$5.46 \times 10^7 - 0.0384 \times 10^7 = 5.4216 \times 10^7$ km.
2	About how many times as far is Mars (farthest) as Mars (closest)? Round to the nearest tenth.	$(4.01 \times 10^8) \div (5.46 \times 10^7) \approx 0.734 \times 10^1 \approx 7.3$ times as far.
3	The spacecraft has a mass of 2.9×10^6 kg and a satellite has a mass of 4.5×10^3 kg. About how many times as heavy is the spacecraft? Write the answer in scientific notation.	$(2.9 \div 4.5) \times 10^3 \approx 0.644 \times 10^3 \approx 6.4 \times 10^2$ times as heavy.

Part 2: Time & Supplies

#	Task	Answer
4	Find the travel time to Mars at its closest in hours (scientific notation) and in days (nearest tenth).	1.365×10^3 hours; $1{,}365 \div 24 \approx 56.9$ days.
5	Find the radio delay to Mars at its closest and at its farthest. Give each in seconds and in minutes (nearest tenth).	Closest: 182 s ≈ 3.0 min. Farthest: $(4.01 \times 10^8) \div (3 \times 10^5) \approx 1.337 \times 10^3$ s ≈ 1,337 s ≈ 22.3 min.
6	A crew of 4 astronauts flies for 60 days. How many calories of food do they need? Write the answer in scientific notation.	$2.5 \times 10^3 \times 4 \times 60 = 600 \times 10^3 = 6 \times 10^5$ calories.

Part 3: Habitat Math

#	Task	Answer
7	Simplify $(3^2)^3 \cdot 3^{-4}$ and evaluate.	$3^6 \cdot 3^{-4} = 3^2 = 9$.
8	A cube module has a volume of 343 m^3. Find its edge. If the edge is doubled, what is the new volume?	Edge $\sqrt[3]{343} = 7$ m. New edge 14 m: $14^3 = 2{,}744$ m^3 ($2^3 = 8$ times as much).
9	Classify each number as rational or irrational: $\sqrt{36}$, $\sqrt{40}$, 0.25, π, $\sqrt[3]{27}$.	Rational: $\sqrt{36} = 6$, 0.25, $\sqrt[3]{27} = 3$. Irrational: $\sqrt{40}$, π.
10	Estimate $\sqrt{40}$ to the nearest tenth without a calculator. Explain.	$6^2 = 36$ and $7^2 = 49$, so $\sqrt{40}$ is between 6 and 7; $6.3^2 = 39.69$ and $6.4^2 = 40.96$, so $\sqrt{40} \approx 6.3$.

Reflection

#	Prompt	Look for
R1	Why does doubling the edge of a cube multiply its volume by 8?	Volume is edge^3, and $(2s)^3 = 2^3 \cdot s^3 = 8s^3$.

#	Prompt	Look for
R2	How do you decide whether a square root is rational?	It is rational exactly when the number is a perfect square (of a rational number); otherwise its decimal never ends or repeats.

Answer Key – Honors

Part 1: Beyond Mars

#	Task	Answer
1	Proxima Centauri is 4.24 light-years away. Find the distance in kilometers in scientific notation (three significant digits).	$9.46 \times 4.24 = 40.1104$, so about 4.01×10^{13} km.
2	How many Earth–Moon distances (3.84×10^5 km) is that? Use scientific notation.	$(4.01 \times 10^{13}) \div (3.84 \times 10^5) \approx 1.04 \times 10^8$ trips.
3	A solar film is 0.00000052 m thick. Write this in scientific notation. How many layers make 1 mm?	5.2×10^{-7} m. 1 mm = 1×10^{-3} m; $(1 \times 10^{-3}) \div (5.2 \times 10^{-7}) \approx 1.92 \times 10^3 \approx 1{,}923$ layers.

Part 2: Speed & Time

#	Task	Answer
4	At 4×10^4 km per hour, how many years would the rocket take to reach Proxima Centauri? (Use 8,760 hours per year.)	$(4.01 \times 10^{13}) \div (4 \times 10^4) \approx 1.00 \times 10^9$ hours; $\div 8{,}760 \approx 1.14 \times 10^5$ years.
5	What speed (km per hour) is needed to reach Mars at its closest in 30 days? Use scientific notation.	30 days = 720 hours; $(5.46 \times 10^7) \div 720 \approx 7.58 \times 10^4$ km per hour.
6	Mission control sends a question to Mars (closest) and waits for the reply. What is the shortest round-trip wait?	$2 \times 182 = 364$ seconds ≈ 6.1 minutes.

Part 3: Habitat Engineering

#	Task	Answer
7	Simplify $(2^5 \cdot 2^{-2}) \div (2^3)^2$ and evaluate.	$2^3 \div 2^6 = 2^{-3} = \frac{1}{8}$.
8	One cube module has a volume of 1,000 m^3. A storage module has 8 times the volume. Find both edge lengths and the ratio of the edges.	$\sqrt[3]{1{,}000} = 10$ m; $\sqrt[3]{8{,}000} = 20$ m. The edges are in the ratio 2 : 1.
9	Show by squaring that $\sqrt{2}$ is between 1.41 and 1.42. Why does its decimal never end or repeat?	$1.41^2 = 1.9881$ and $1.42^2 = 2.0164$, so $1.41 < \sqrt{2} < 1.42$. $\sqrt{2}$ is irrational, so its decimal is non-terminating and non-repeating.
10	The food budget is 1×10^7 calories. Choose a crew size and find the longest mission it supports at 2.5×10^3 calories per person per day.	Answers vary. Example: 4 astronauts use 1×10^4 calories per day, so $(1 \times 10^7) \div (1 \times 10^4) = 1{,}000$ days.

Reflection

#	Prompt	Look for
R1	Explain why $a^{-n} = \frac{1}{a^n}$ using a pattern of powers.	Each step down in the exponent divides by a: a^2 , a^1, $a^0 = 1$, $a^{-1} = \frac{1}{a}$, $a^{-2} = \frac{1}{a^2}$.
R2	Why can the product of two irrational numbers be rational? Give an example.	$\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$; the irrational parts can combine into a perfect square.